CLAIMS

What is claimed is:

1. A system for imaging an interior of a target space, comprising:

- a first mobile unit equipped with a directional signal transmitter configured to emit a signal capable of penetrating the boundaries of the target space;
- a second mobile unit equipped with a directional signal receiver configured to detect said signal after it has passed through the target space;
- a synchronization module configured to coordinate the timing and positioning of the first and second mobile units; and
- a processing system configured to receive data from the signal receiver and to computationally reconstruct an image of objects within the target space by analyzing the attenuation of the signal.

2. The system of claim 1, wherein the first and second mobile units are vehicles.

3. The system of claim 1, wherein the signal is a non-ionizing radiation signal.

4. The system of claim 3, wherein the signal is a millimeter-wave signal.

5. The system of claim 1, wherein the processing system is further configured to generate a three-dimensional tomographic image by combining data from multiple projections captured as the mobile units move in a synchronized path relative to the target space.

6. A method for imaging an interior of a target space, comprising the steps of:

- positioning a first mobile transmitter unit and a second mobile receiver unit on substantially opposite sides of the target space;
- transmitting, from the first unit, a penetrating signal through the target space towards the second unit;

- receiving, at the second unit, the signal after it has been attenuated by the target space and its contents;
- communicating the received signal data to a processing system; and
- computationally generating an image of the contents of the target space based on a differential analysis of the signal's attenuation.

7. The method of claim 6, wherein the penetrating signal is a millimeter-wave signal.

8. The method of claim 6, further comprising the step of moving the first and second units in a synchronized path relative to the target space to capture a plurality of projections for tomographic reconstruction.